

# Adjustment Margins and Tax–Benefit Cushioning of Earnings Losses in the United Kingdom\*

Vahid Ahmadi<sup>†</sup>

August 2026

## Abstract

How much of an earnings loss the UK tax–benefit system absorbs depends on how the loss is distributed across workers and labour-market states, not only on its size. Using the 2024–25 Family Resources Survey and PolicyEngine UK, I hold an aggregate employee-earnings loss fixed—calibrated, as an application, from the 2025 US tariff schedule through a declared accounting bridge, not an estimated first stage—and vary its incidence factorially: diffuse wage cuts across all exposed workers, concentrated wage cuts imposing the displacement draw’s worker-level losses without job loss, and displacement of the same workers. Diffuse cuts are cushioned at 43.9 per cent, displacement at  $34.0 \pm 4.7$ : a 10.0-point gap, of which 9.9 points reflect loss concentration—which exhausts marginal-rate tax relief—and only 0.1 points the employment state itself. Record-resampling, leave-one-record-out and leave-one-sector-out audits bound the contrast without significance claims; sensitivity analysis varies pensions, take-up, duration equivalents, adjustment shares and worker selection.

**Keywords:** automatic stabilisers, tariffs, trade shocks, microsimulation, poverty, inequality, public finances

**JEL codes:** F13, F16, H23, H24, I38, J65

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\* **Acknowledgements:** I thank Max Ghenis and Seyed Mahdi Hosseini for valuable comments and helpful discussions.

<sup>†</sup>Research Associate at PolicyEngine. Email: [vahid@policyengine.org](mailto:vahid@policyengine.org).

# 1 Introduction

Who bears an earnings loss when the same aggregate amount can be distributed through different workers and labour-market states? This paper answers that question for the United Kingdom. I hold an aggregate loss in employee earnings—roughly £886 million a year in the application, calibrated from the 2025 US tariff schedule—fixed, and compare its fiscal and distributional consequences across a factorial family of incidence patterns: diffuse proportional wage cuts across all exposed workers, concentrated wage cuts imposing the displacement draw’s exact worker-level losses without any job loss, displacement of the same selected workers into unemployment, and mixed transition paths with re-employment penalties. Defining the cushioning rate as  $1 - \Delta Y / \Delta E$ , where  $\Delta E$  is the gross employee-earnings loss and  $\Delta Y$  the household disposable-income loss, diffuse wage cuts are cushioned at 43.9 per cent while displacement is cushioned at  $34.0 \pm 4.7$  per cent—a gap of 10.0 percentage points. The two headline scenarios differ in more than the adjustment margin: they also concentrate the loss on different workers. The factorial middle cell separates the two channels, and the answer is one-sided. Concentrating the same aggregate loss on the displacement draw’s workers, while keeping everyone employed, accounts for 9.9 points of the gap; moving those same workers, bearing the same worker-level losses, from employment into unemployment accounts for only the remaining 0.1 points (Section 4.3). The mechanism is therefore a loss-size effect inside the statutory schedule: a diffuse in-work reduction is offset automatically at marginal income-tax and National Insurance rates, whereas a complete earnings loss exhausts that relief and lands on means-tested Universal Credit—whose statutory entitlement responds to income, not employment status, so the state change itself adds almost nothing *within the modelled annual system*. The institutions that do condition on employment status—New Style JSA, monthly assessment periods, waiting and claiming frictions, work-search conditionality—are exactly the ones outside the model, so the near-zero state component is a statement about the modelled statutory rules, not about the real-world experience of job loss. Actual receipt additionally depends on capital holdings and claiming behaviour that the survey data only partially observe. The gap is not an artefact of the HBAI treatment of pension saving: recomputed under a pension-gross income concept, both margins’ cushioning falls by roughly four percentage points but the wage-minus-displacement contrast is essentially unchanged (Section 4.2).

The comparison matters because the UK tax–benefit system is nonlinear. For an equal aggregate loss, a broad reduction in earnings leaves most households in work and inside the tax schedule, whereas a concentrated loss moves households across benefit-eligibility thresholds. Fiscal incidence therefore depends on who loses earnings, whether the loss is temporary, and whether employment resumes—and a single wage-cut or displacement scenario should not be read as a forecast of realised trade adjustment. The two poles also price a structural disagreement: flexible-wage models turn a tariff primitive into a wage shock (Ignatenko et al., 2025), while downward-nominal-rigidity models turn the same primitive into a job shock (Rodríguez-Clare et al., 2025). Rather than endogenising the margin, I hold the aggregate loss fixed and report what each pole implies for households and the Exchequer.

The epistemic status of every input is stated once, here, and in the design table of Section 3.1, rather than repeated throughout. The estimand is conditional: the statutory tax–benefit and distributional response to a specified earnings-loss path, not a causal estimate of the 2025 tariff episode. The 2025 US tariff schedule supplies sector weights and shock magnitudes through a transparent accounting bridge (tariff rate  $\times$  US-export intensity  $\times$  a demand calibration  $\times$  a wage-bill-incidence assumption); the bridge deliberately omits prices, quantities, imported in-

puts, productivity, profits, and firm entry. Reported  $\pm$  values are assignment dispersion from the Monte Carlo allocation of displacement, not standard errors; sensitivity of the primary contrast to which FRS records were sampled is scored separately by a household record-resampling exercise and a leave-one-record-out audit, neither of which is a design-based confidence interval. Mixed-path shares are disciplined by external evidence but not estimated.

One negative result should be on the table from the outset. A monthly HMRC product–destination panel confirms a post-policy decline in US-bound exports, but at sector rank the calibration fails its own falsification test: the rank correlation between predicted and realised sector export falls among the ten largest exposed divisions is zero (Section 4.4). This failure bounds the paper’s claims—the sector calibration must not be read as measured tariff incidence—but it does not touch the cushioning comparison, which conditions on a declared loss rather than on the bridge. The leave-one-division-out audit across all 23 exposed divisions shows the wage-minus-displacement contrast surviving every rerun (range 7.6–11.3 points): the calibration determines *who* is shocked, not the statutory asymmetry between intensive and extensive relief.

The second stage uses the 2024–25 Family Resources Survey and PolicyEngine UK to compute taxes, National Insurance, benefits, household disposable income, poverty, inequality, and Exchequer effects. I report a common aggregate loss under alternative incidence patterns, decompose the result by tax and benefit channel, and show sensitivity to spell duration, worker selection, benefit take-up, and re-employment. Between the polar bounds a transition surface varies the loss share borne by workers who re-enter services jobs after a lag; its interior points rest on assumed shares, lags and a cross-sectional earnings penalty, so the response surface, not any single interior calibration, is the reported object (Section 4.2).

The paper makes three contributions. First, it provides a reproducible UK microsimulation framework for mapping earnings losses—here calibrated to trade exposure as an application—into fiscal and distributional outcomes. Second, it separates, factorially, how much of the wage-cut versus displacement cushioning gap reflects concentrating a fixed loss on selected workers and how much reflects the change in employment state itself. Third, it quantifies which statutory channels drive the differences and which assumptions—timing, take-up, pensions, and worker selection—are most important for policy conclusions. The remainder of the paper describes the evidence and scenario design, presents the polar bounds and mixed paths, reports policy counterfactuals, and discusses limitations and reproducibility.

## 2 Literature

This paper sits at the intersection of four literatures: the incidence of trade shocks; the evidence on how workers adjust to them; the institutional analyses of the 2025 US tariffs on the UK; and the microsimulation literature on tax–benefit cushioning of earnings shocks. Existing macro–micro studies impose labour-market shocks on tax–benefit models, while tariff studies map price and income exposure into heterogeneous household welfare—the World Bank’s Household Impacts of Tariffs framework combines household consumption and income portfolios (Artuc et al., 2021), and the OECD links its METRO trade model to household-budget surveys (Luu et al., 2020). The narrower contribution here is a UK application that maps the 2025 partner-country export-demand exposure into alternative labour-income stress scenarios and then through the statutory tax–benefit system; relative to those economy-wide tools it supplies only the terminal tax–benefit module, conditioning on an imposed labour-income distribution rather than deriving it from production (see Section 3.1).

## 2.1 Trade shocks and their incidence

Two recent papers anchor the structural end and frame the adjustment-margin question as its poles. [Ignatenko et al. \(2025\)](#) build a quantitative general-equilibrium model of the 2025 tariffs in which wages are flexible and labour supply elastic, so employment is an equilibrium outcome whose sign flips with revenue recycling; their replication data provide the UK-specific aggregate anchors I use. [Atkin et al. \(2025\)](#) instead apply a wedge framework to Chilean administrative data with factor supplies fixed: all adjustment runs through flexible wages and prices, and their “statutory versus economic incidence” framing is the one I adopt. Between the poles, [Rodríguez-Clare et al. \(2025\)](#) embed the 2025 trade war in a model with downward nominal wage rigidity, in which rigid wages turn the same tariff primitive into an employment shock that flexible-wage models turn into a wage shock—precisely the disagreement my adjustment-margin axis makes an explicit scenario dimension. [Michelena et al. \(2025\)](#) find 2025 employment and export losses concentrated in the targeted countries, and [den Besten and Känzig \(2025\)](#) show, in a narrative series over 1840–2024, that tariff increases are robustly contractionary—discipline for treating the shock as a negative export-demand shock.

The reduced-form lineage begins with [Autor et al. \(2013\)](#): import competition caused large, persistent employment and wage losses in exposed US commuting zones (effects persisting to 2019; [Autor et al., 2021](#)). [Caliendo et al. \(2019\)](#) provide the structural counterpart, [Adão et al. \(2023\)](#) the mapping from regional estimates to aggregates, [Kim and Vogel \(2021\)](#) the money-metric worker welfare translation with regressive incidence, and [Galle et al. \(2023\)](#) the coexistence of aggregate gains with group-specific losses. My contribution relative to this work is to replace stylised worker groups and transfers with real households and the actual UK tax–benefit system. The 2018–19 US episode supplies the sharpest evidence on who pays a tariff: pass-through into US import prices was essentially complete ([Amiti et al., 2019](#); [Fajgelbaum et al., 2020](#)), retail prices adjusted more slowly ([Cavallo et al., 2021, 2025](#)), and US manufacturing employment fell on net ([Flaen and Pierce, 2019](#)). The retaliation side is the closest analogue to my channel—a partner’s tariff arriving as an export-demand shock ([Vaugh, 2019](#); [Blanchard et al., 2019](#)). The broader distributional literature establishes redistribution along both consumption and income margins ([Fajgelbaum and Khandelwal, 2016, 2022](#); [Borusyak and Jaravel, 2021](#)). Two propagation channels my direct-channel design excludes have well-measured analogues: supply-chain transmission ([Barrot and Sauvagnat, 2016](#); [Acemoglu et al., 2016](#)) and local-services spillovers ([Moretti, 2010](#); [Gathmann et al., 2020](#)); Section 6 carries the caveat. Exchange rates absorb part of a tariff, imperfectly and state-dependently ([Jeanne and Son, 2024](#); [Corsetti et al., 2025](#); [Kalemli-Özcan et al., 2026](#); [Auray et al., 2020](#)); I hold the exchange rate fixed. Early assessments of the 2025 wave again find the incidence landing on the tariff-imposing country’s consumers ([Fajgelbaum and Khandelwal, 2026](#))—complementary to my question, the incidence on the tariffed *partner’s* workers.

## 2.2 Adjustment margins: how workers absorb trade shocks

The worker-level evidence says the margin is mostly earnings, not permanent unemployment. [Autor et al. \(2014\)](#) find exposed US workers lose earnings amid churn rather than indefinite joblessness; [Traiberman \(2019\)](#) shows occupation-specific human capital produces reallocation with persistent wage losses. Because my shock is an export-demand shock rather than import competition, the most direct precedents are [Hummels et al. \(2014\)](#) and [Garin and Silvério \(2024\)](#), who show exporters pass export-demand shocks into incumbent wages—evidence for including

a wage-cut dimension. The rent-sharing literature disciplines its plausible scale: elasticities of wages to firm-specific rents of 0.05–0.15 (Card et al., 2018; Kline et al., 2019). Dauth et al. (2021) provide the cleanest evidence for the reallocation family, Dix-Carneiro (2014) the adjustment-cost horizons and age gradients, and Rud et al. (2022) decompose displacement costs into wage cuts on re-employment versus non-employment spells—directly calibrating the split between my families.

The UK’s own history points to a third margin: benefit-financed economic inactivity. Beatty et al. (2007) document that coalfield job losses became male inactivity on incapacity benefits, and Beatty and Fothergill (2017) show that 1980s–90s industrial job destruction still inflates the deficit—the direct UK precedent for my fiscal-incidence question, retrospective and area-level where mine is prospective and household-level. The margin is live: disability claims still respond to local slack (Roberts and Taylor, 2022), hidden unemployment persists (Beatty and Fothergill, 2023), and working-age health-related benefit rolls reached 4.2 million by 2024 (IFS, 2024). For UK-specific elasticities, De Lyon and Pessoa (2021) show workers in Chinese-import-competing industries lost employment years and earnings (see also Dorn and Levell, 2024; Aragón et al., 2018). Closest to my object, Irastorza-Fadrique et al. (2025) study UK household responses to trade shocks—added-worker effects, retirement timing, self-employment—but model behaviour, not tax–benefit mechanics; I treat their margins as complementary.

### 2.3 The 2025 tariffs and the UK

The institutional literature on the 2025 shock is rich but stops above the household. The OBR puts the permanent UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent (OBR, 2025); NIESR provides NiGEM scenarios (NIESR, 2024). The Bank of England estimates the effective US tariff on the UK at roughly 9 per cent and finds trade diversion has mildly *lowered* UK import prices (Bank of England, 2025; Dhingra, 2025)—which licenses isolating the export-demand/earnings channel while holding price and monetary channels fixed. The IMF and WTO frame the aggregates (IMF, 2025; WTO, 2025); UK firm surveys find mostly small expected effects, with the US taking about 22 per cent of UK exports (CEP, 2025). The geography is sharp: Coventry sends 22 per cent of its exports to the US (Centre for Cities, 2025); auto tariffs are estimated to cost £10,000 million of GDP over five years, 62 per cent borne by the West Midlands (City-REDI, 2025); SMMT data show US vehicle exports collapsing between April and May 2025 (SMMT, 2025). Some 80 per cent of UK steel exports go to Europe—the basis of my steel caveat—and the December 2025 pharmaceutical deal was paid for through VPAG rebate cuts and NHS pricing concessions, making pharma a fiscal/pricing shock rather than an employment shock (ABPI, 2025). The Resolution Foundation’s household-facing analysis is descriptive rather than microsimulated (Resolution Foundation, 2025a), and the IFS has no dedicated distributional publication on the 2025 tariffs; that gap defines the contribution. For UK trade-shock precedents, Dhingra et al. (2017) map Brexit trade costs to incomes, Fetzer and Wang (2024) provide district-level incidence, Fetzer (2019) and Bloom et al. (2019) the political-economy and uncertainty channels, and Breinlich et al. (2022) and Levell et al. (2017) benchmark the consumption-price channel that this cash-income exercise holds fixed.

## 2.4 Tax–benefit cushioning of earnings shocks

The methodological ancestors are [Dolls et al. \(2012\)](#), who show EU tax–benefit systems absorb about 38 per cent of a proportional income shock and 47 per cent of an unemployment shock. Their *UK* cells are the right comparator: a UK income-shock stabilisation coefficient of 0.352 (income tax 0.267, social insurance 0.054, benefits 0.031) and a UK unemployment-shock coefficient of 0.415 (tax 0.191, SIC 0.061, out-of-work benefits 0.163). My 43.9 per cent wage-cut and  $34.0 \pm 4.7$  per cent displacement rates are of a similar order; Section 4.2 focuses on the *composition* of the total for the exposed-worker population. The marginal-rate framing of the income tax as an automatic stabiliser originates with [Auerbach and Feenberg \(2000\)](#) and is formalised by [McKay and Reis \(2016\)](#). [Sutherland and Figari \(2013\)](#) establish EUROMOD’s stress-test use; my direct methodological sibling is [Brewer and Tasseva \(2021\)](#), who pass the Covid-19 labour-market shock through UKMOD—the same architecture, applied here to a trade-policy shock without the emergency schemes and with the adjustment margin as the central axis. The macro–micro linkage literature supplies the shock-derivation lineage ([Bourguignon et al., 2008](#); [Peichl, 2016](#); [Navicke et al., 2013](#)). On the US side, exposed commuting zones saw rising disability, retirement and welfare payments offsetting only a small fraction of losses ([Autor et al., 2013](#)), and [Hyman et al. \(2024\)](#) show wage insurance raises re-employment—a candidate counterfactual reform for this framework. The nearest tariff-microsimulation links are US consumption-side models ([Tax Policy Center, 2025](#); [Budget Lab, 2025](#); [Acosta and Cox, 2024](#)). Two institutional facts motivate the cushioning asymmetry examined here: UC’s capital rules—thresholds frozen in cash terms since 2006—reduced or extinguished entitlement for around two million income-eligible families ([Resolution Foundation, 2025b](#)), and take-up of income-related benefits is well below 100 per cent ([DWP, 2024](#)), so statutory replacement rates can overstate what displaced workers receive. Take-up is examined as a headline sensitivity (Sections 3.6 and 4.2); the capital rules are a real-world gate the plain FRS data cannot exercise (Section 6).

## 3 Methodology

### 3.1 Design: primitives first

Throughout, “derived” means generated by the stated scenario mapping, not causally estimated. The central demand parameter is a transparent unit stress,  $\varepsilon = 1.0$ , rather than an estimate. The grid includes  $\varepsilon = 0.4$  as an OBR-style lower anchor, 2.0 as the former April-outturn high case, and 3.0 as a severe sensitivity. The April movement is not used to identify the centre because it also reflects anticipation, front-running, prices, exchange rates, composition and other demand conditions. Likewise,  $\pi = 1.0$  is a deliberately strong full wage-bill-incidence assumption. The parameter grid is scenario sensitivity, not a statistical confidence interval.

A reader is entitled to ask, before any number is reported, which parts of this design are read off data and which are assumed. The answer, stated in full:

| Object                    | Status                                                                                                                                                                                     |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tariff schedule $\tau_j$  | <b>Data</b> (policy documents: the April 2025 schedule, the May 2025 EPD, the December 2025 pharmaceutical agreement).                                                                     |
| US-export intensity $x_j$ | <b>Data</b> (HMRC RTS/OTS customs exports over ONS ABS division turnover; Appendix A).                                                                                                     |
| Export fall               | <b>Data</b> in the measured family (realised HMRC outturn, May 2025–February 2026); <b>assumed</b> in the tariff families through declared low, central, high and severe demand scenarios. |
| Pass-through $\pi$        | <b>Assumed</b> ( $\pi = 1.0$ ), varied on a grid (Appendix A).                                                                                                                             |
| Adjustment margin         | <b>Assumed</b> , and deliberately varied. This is the paper’s free dimension.                                                                                                              |
| Household incidence       | <b>Computed</b> from statutory tax and benefit rules in PolicyEngine UK; no behavioural or reduced-form step.                                                                              |

The margin is assumed because the literature does not agree on it, and the disagreement is structural rather than empirical: Ignatenko et al. (2025) derive the employment response from elastic labour supply and flexible wages, while Rodríguez-Clare et al. (2025) derive it from downward nominal wage rigidity, so the same tariff primitive arrives as a wage shock in the first model and a job shock in the second. I do not endogenise the margin. I price the disagreement—holding the aggregate earnings loss fixed and reporting what each pole implies for households and the Exchequer. That is a weaker claim than a structural model makes, and a more auditable one.

The shock enters through primitives, not through an assumed employment effect. The pipeline is:

$$s_j = \tau_j x_j \varepsilon \pi, \quad (1)$$

where  $s_j$  is an imposed employee-wage-bill stress for SIC 2007 division  $j$ ,  $\tau_j$  is its tariff rate,  $x_j$  is US-export intensity,  $\varepsilon$  is an export-demand calibration and  $\pi$  is a *wage-bill-incidence assumption*. Calling  $\pi$  pass-through would conflate distinct mechanisms. A structural transmission would separately map the tariff into export prices and quantities, quantities into domestic production and value added, and production into factor demand, productivity, profits, wages, hours and employment. Equation (1) compresses those links into one accounting bridge. It neither estimates a production function nor identifies the causal response of productivity, output or labour demand.

$x_j$  is US goods exports as a share of division turnover, built from HMRC trade data and ONS Annual Business Survey turnover denominators; Appendix A documents the frozen provenance and denominator choice. The central  $\varepsilon = 1.0$  is a transparent unit stress, not an elasticity estimate; it amounts to a unit tariff-to-export-demand semi-elasticity, under which each percentage point of tariff removes one per cent of US-bound demand, and  $\varepsilon = 0.4$  is the lower anchor, in the spirit of the trade-elasticity assumptions underlying the OBR’s March 2025 tariff scenarios (OBR, 2025). The central  $\pi = 1$  assigns the full calibrated exposure to the employee wage bill; it is intentionally severe and is not established by the economics literature. The sensitivity grid varies  $\varepsilon\pi$ , but this is parameter sensitivity, not an uncertainty interval around an identified coefficient. The implied gross earnings loss is therefore a conditional scenario input. OECD METRO and the UK Trade Modelling Review instead place tariff shocks inside multi-sector production systems in which prices, output, value added and factor demand adjust jointly (Luu et al., 2020; Department for International Trade, 2021); the World Bank HIT framework similarly labels its direct household calculations first-order and excludes productivity and substitution responses

(Artuc et al., 2021). This paper begins only after those missing production-side mechanisms would have generated a labour-income distribution.

### 3.2 Public product–destination benchmark

A monthly HMRC OTS panel (January 2018–May 2026; US, Canada, Japan, Australia) tests whether post-policy US-bound exports move differently from the same HS4 products sent elsewhere. The design regresses a log destination gap on a post-May-2025 indicator with product and seasonal effects and a linear trend, weighting by capped pre-policy US export values, and reports control-destination, weighting, sample-start and placebo-date checks. It controls for product shocks common across destinations but not US-specific demand or other coincident policies, and it is not used to estimate the demand or incidence parameters of Equation (1); its role is falsification. The full design and all estimates are in the Online Supplement (Sections S1–S2).

### 3.3 The tariff schedule and the EPD counterfactual

Two tariff scenarios bracket the policy. *Full tariffs*: the April 2025 schedule—10 per cent baseline on UK goods, 25 per cent on motor vehicles (SIC 29) and steel (SIC 24). *EPD*: the deal-mitigated schedule—autos at the 10 per cent in-quota rate on up to 100,000 vehicles; steel with conditional, partial relief; pharmaceuticals (SIC 21) exempt under the December 2025 agreement. The automotive scenario applies the 10 per cent rate to the modelled 2024 exposure because the quota is approximately equal to that year’s US-bound volume; it does not model quota allocation, above-quota growth or an endogenous volume response around the ceiling. The difference between the two scenarios, run through identical downstream machinery, prices the EPD for households and the Exchequer. Two caveats travel with the schedule. Steel: about 80 per cent of UK steel exports go to the EU, so the US tariff is a minority driver of the sector’s exposure and the steel cell should be read alongside EU/UK safeguards. Pharmaceuticals: the exemption was purchased through VPAG rebate cuts and NHS pricing concessions, so pharma’s incidence is fiscal and price-side, not an employment shock; I model its employment shock as zero under the EPD and discuss the fiscal channel separately (Section 5).

### 3.4 Adjustment-margin families

The derived  $s_j$  is realised on Family Resources Survey 2024–25 employees in exposed divisions. The submission design treats the mixed adjustment surface as the conceptual centre: statutory cushioning is evaluated for a common expected sector wage-bill loss while the share delivered through temporary non-employment, survivor earnings reductions, and re-employment is varied. The pure displacement and proportional wage-cut cases are transparent bounds on that surface. Inactivity, literal sectoral reallocation and the EPD schedule are complementary sensitivities, not alternative causal estimates. Wage cuts remove  $\sum_j s_j B_j$  deterministically; displacement removes that amount in expectation under the risk model and is numerically integrated using the balanced design below.

**Displacement (ADH short run).** Within each exposed division every employee has baseline risk  $s_j$ , irrespective of the record’s survey weight. The primary estimator generates repeated random-start systematic candidates, ordered across age and within-industry earnings ranks, and

retains the candidate that best matches the expected industry wage-bill and weighted-headcount losses while secondarily balancing broad age and earnings margins. This is balanced numerical integration, not a survey estimator: conditioning on aggregate balance intentionally changes exact record-level marginal inclusion probabilities, and because the retained candidate minimises a score dominated by the total wage-bill target, per-division losses are matched softly rather than exactly and best-of-candidates selection carries no formal unbiasedness guarantee. The claim that the balanced design removes the declared amount in expectation is therefore established empirically, by its agreement with the unbiased comparator, not by construction. Selected workers move to unemployment with earnings, hours, pension contributions and statutory pay zeroed. An independent Bernoulli estimator, which preserves the declared first-order probabilities exactly but has much larger allocation dispersion, is reported as the unbiased assignment-rule comparator; at the unit scale the two agree on cushioning to half a percentage point (Section 4.2).

**Concentrated wage cut (factorial middle cell).** The headline wage-cut versus displacement contrast changes two things at once: the adjustment margin and the concentration and identity of the losers. To separate them, a third family imposes the displacement draw’s exact worker-level losses—the same balanced draw on the same seeds, each drawn worker losing their entire annual earnings, with pension contributions and statutory pay scaling to zero exactly as for any in-work earnings change—while leaving every worker employed at baseline hours. Comparing the diffuse wage cut with this cell isolates concentration and worker selection at fixed employment state; comparing this cell with displacement isolates the employment-state change at fixed workers and worker-level losses. Because employment is binary, a “diffuse displacement” cell is undefined, so the sequential path diffuse  $\rightarrow$  concentrated  $\rightarrow$  displaced is the unique monotone decomposition of the gap and coincides with the Shapley allocation over the two factors (Section 4.3).

**Wage cuts with reallocation (Dauth/Traiberman long run).** No job loss; every employee in division  $j$  takes a proportional cut of  $s_j$ , so the weighted aggregate earnings removed equals  $\sum_j s_j B_j$  by construction—exact earnings-equivalence with the displacement family’s expected removal. Employee pension and salary-sacrifice contributions are scaled proportionally with the cut, mirroring their zeroing under displacement; this lowers taxable pay (and hence tax relief) but also reduces the household’s pension outflow, so under the HBAI disposable-income concept part of the earnings loss is absorbed by reduced retirement saving rather than by current income.

**Inactivity (Beatty–Fothergill UK history).** The displacement draw, but displaced workers aged 50 and over exit to economic inactivity, the benefit-financed route of UK deindustrialisation, rather than unemployment. The route is benefit-financed in the model, not just relabelled: every inactive worker is flagged as having limited capability for work-related activity, so their household’s Universal Credit includes the LCWRA health element (£217.26 a month in 2026—the reduced rate applying to new claims from April 2026 under the Universal Credit Act 2025; existing claimants retain the higher pre-reform rate, and all modelled inactivity claims are new claims). In policyengine-uk 2.89.2 the survey-imputed `uc_LCWRA_element` is stored, so the post-shock benefit-unit value is set to the maximum of the stored amount and one annual statutory element; an existing recipient is never paid a second element, unaffected units are unchanged,

and the run hard-errors if the element fails to reach a newly flagged person’s benefit unit. This mirrors the [Beatty et al. \(2007\)](#) route by which industrial job loss became incapacity-benefit inactivity. The strong assumption should be stated plainly: *all* displaced workers aged 50 and over are assumed to pass the Work Capability Assessment, so the family is an upper bound on the health-element route; no partial-qualification sensitivity (for example, 50 per cent of the older displaced qualifying) has been simulated, which is a limitation of the present scenario set. Work-search conditionality carries no separate fiscal machinery in the model, so the LCWRA element is the entire modelled wedge between inactivity and unemployment.

**Reallocation into services (ADH/Autor et al., 2014).** The same displacement draw on the same seed—so the two families are paired worker-for-worker—but the drawn workers are re-employed rather than left out of work. Their `sic_industry_division` is reassigned to one of four services destinations (SIC 47 retail, 86 human health, 49 land transport, 56 food and beverage service) in proportion to those divisions’ FRS employee-weight shares (29.5, 45.8, 9.2 and 15.5 per cent), and their annual earnings are cut by a penalty calibrated on the FRS itself: weighted mean annual employee earnings of £48,272 over 2,047 hours in the exposed goods divisions against £34,594 over 1,701 hours in the four destinations, a **28.3 per cent annual-earnings penalty** that embeds the hours fall. Two variants are reported: an hours- and age-controlled **low-penalty** case of 14.0 per cent (the pure hourly-wage gap), and a **three-month lag** in which workers are out of work for a quarter of the year before the services job starts, scaling annual earnings and hours by a further  $1 - 3/12$ . ASHE 2024 full-time medians bracket the hours-controlled figure at 10–20 per cent; the FRS all-employee penalty is larger because it also captures the shift into part-time work. Because this family removes only the penalty on the movers rather than the full earnings-equivalent aggregate, its pound figures are comparable with the displacement family (identical worker sets, paired draws) but *not* with the wage-cut family; its cushioning *rate* is comparable with both. Appendix A reports the results and mechanics.

**Common-loss transition path.** The redesigned transition margin combines a transition-equivalent annual earnings/hours penalty, service-sector re-employment and survivor earnings reductions while preserving the common-loss normalization. Let  $s$  be the sector shock,  $\lambda$  the share of the declared loss assigned to transition workers, and  $d = 1 - (1 - r)(1 - \ell/12)$  the annual earnings loss of a transition worker, where  $r$  is the re-employment penalty and  $\ell$  the lag in months. Transition probability is  $p = \lambda s/d$ ; survivors take a cut  $c = (1 - \lambda)s/(1 - p)$ . Hence  $pd + (1 - p)c = s$  in expectation. The illustrative interior calibration uses  $\lambda = 0.70$ ,  $r = 0.283$  and  $\ell = 3$ ; the transition share is bounded by the displacement-cost decompositions of [Rud et al. \(2022\)](#) and [Dix-Carneiro \(2014\)](#), which attribute the majority of displacement costs to non-employment spells and re-employment penalties rather than survivor wages, but it is an assumption, not an estimate. It is one point on the response surface, which remains the paper’s central object. In the annual simulation, the lag is represented by an earnings/hours factor: workers remain employed for PolicyEngine purposes, so no within-year unemployment or JSA/UC spell is generated. This is an accounting normalization for comparing paths, not an estimate of the probability of re-employment or the UK tariff response. The 14 per cent hours/age-controlled penalty and zero-, three- and six-month lags are sensitivity dimensions.

**Mixed adjustment and factorial scenario testing.** Let  $\lambda \in [0, 1]$  denote the share of the sector shock delivered through temporary non-employment rather than an incumbent earnings reduction. A worker with sector shock  $s$  is displaced with probability  $p = \lambda s$ . Conditional on surviving, the proportional wage cut is

$$c = \frac{s - p}{1 - p},$$

so expected earnings loss is exactly  $p + (1 - p)c = s$  for every worker. The endpoints reproduce pure wage cuts ( $\lambda = 0$ ) and pure temporary displacement ( $\lambda = 1$ ); because the mixed family draws displacement independently, the  $\lambda = 1$  endpoint reproduces the Bernoulli comparator rather than the balanced primary estimator. A separate reallocation family moves displaced workers into observed service destinations, with either an immediate earnings penalty or a three-month non-employment lag. It is paired worker-for-worker with displacement but does not preserve the common aggregate loss unless its penalty is explicitly re-scaled, so its pound totals are reported as a transition sensitivity rather than combined with the mixed grid. The exploratory grid crosses  $\lambda \in \{0, .25, .5, .75, 1\}$  with export-demand calibrations  $\varepsilon \in \{0.4, 1, 2, 3\}$  on five common assignments per cell. This follows the companion UK AI study’s factorial principle: vary shock magnitude separately from adjustment composition. It is scenario testing, not a probability distribution over future outcomes.

**The wage-cut margin as an upper bound, and a literature-disciplined mixed sensitivity.** The pure wage-cut family should be read against the rent-sharing evidence, because it embeds a survivor-wage response far above anything estimated. Delivering the entire sector wage-bill loss through proportional cuts to incumbent workers’ wages imposes a unit wage response to the sector exposure index. [Garin and Silv rio \(2024\)](#) estimate an incumbent-wage elasticity of roughly 0.13–0.14 to firm-specific output shocks in Portuguese matched data; the rent-sharing survey of [Card et al. \(2018\)](#) places typical elasticities at 0.05–0.15; and [Hummels et al. \(2014\)](#) find small within-job wage responses in Danish data alongside losses through displacement. The wage-cut family ( $\lambda = 0$ ) is therefore an *upper-bound stress scenario for survivor-wage incidence*, retained because it prices the flexible-wage pole of the structural disagreement.

The scenario surface also contains a 15/85 point ( $\lambda = 0.85$ ): 15 per cent of the imposed wage-bill loss delivered through survivor wage cuts, disciplined by the upper end of the cited wage-response estimates. A wage elasticity to firm output or rents is not algebraically the share of an aggregate wage-bill loss borne by survivor wages—identifying that share requires linked employer–worker data—so this is one surface point among others, subordinate to the transition path above as the paper’s single illustrative interior calibration.

Two scope limits on the family set should be stated here rather than in a footnote. The wage-cut family models the *earnings consequence* of reallocation without moving anyone between sectors: workers keep their observed industry and hours, and the proportional cut stands in for the wage penalty that a completed reallocation would carry; the reallocation family does move them, but who reallocates is assumed (it is the displacement draw) rather than estimated. And no household-side behavioural margin is active—added-worker responses, earlier or later retirement, and shifts into self-employment ([Irastorza-Fadrique et al., 2025](#)) are all suppressed, as are any labour-supply responses to the tax and benefit changes themselves. All four families are therefore first-round, rules-based incidence.

A third caveat concerns selection within divisions. Export-exposed workers are unlikely to be representative of their division: exporters pay a wage premium and pass export-demand shocks

into incumbent wages (Hummels et al., 2014; Garin and Silvério, 2024). Cushioning is nonlinear in earnings, so uniform risk may misstate its tax/benefit composition. The longitudinal-LFS cell, earnings-band and QRF sensitivities vary observed risk shape while recalibrating each division to its wage-bill target. They quantify selection- model sensitivity but cannot identify exporter status or tariff-specific worker risk.

The FRS support for the transition is thin. Of 34,966 person records, 12,800 have employee earnings and 1,018 are in mapped exposed divisions, but the reference probabilities imply only 9.7 selected records. The largest exposed record has a person weight above 19 thousand. Balanced integration constrains the realised aggregate and observed composition; it does not create survey information or establish population representativeness. I therefore report the affected record count, Kish-style effective number of gross-loss contributions, largest-record loss share, and balanced-versus-Bernoulli mean and SD comparisons. Subgroup rankings remain exploratory.

**Duration-equivalent annual stresses.** The annual PolicyEngine model cannot represent a worker employed for part of a year and unemployed for the remainder with internally consistent monthly UC assessment, work status and earnings. I therefore do not simulate three- or six-month unemployment spells. Instead I scale  $s_j$  by  $d/12$ , for  $d \in \{3, 6, 12\}$ , and assign a correspondingly smaller number of coherent full-year displaced states. These rows preserve the expected worker-month earnings loss associated with each duration while changing the extensive-margin stock. They are labelled *duration-equivalent annual stresses*, not spell estimates. Wage-cut rows use the same scale, preserving the common- loss comparison at every duration and at both the OBR-style 0.4 and unit anchors. Appendix E bounds the bias from this annualisation with a parameter-based comparison of the duration-equivalent stress against a correct monthly UC assessment for a representative displaced worker: UC entitlement per worker-month is identical by construction, the primary 12-month contrast is unaffected, and the residual differences at 3- and 6-month scales run through partial-year income tax and payment timing.

The margin is the paper’s central axis because the tax–benefit system treats the two poles differently: an in-work earnings reduction is offset automatically at marginal tax and taper rates, while a complete loss exhausts marginal-rate relief and lands on the means-tested part of the system. Whether that difference is driven by the employment-state change or by the concentration of the losses is precisely what the factorial middle cell measures rather than assumes (Section 4.3).

### 3.5 Longitudinal LFS imputation benchmark

Because uniform within-division assignment suppresses observable worker heterogeneity, longitudinal LFS Study 9490 supplies cell, earnings-band and regularised quantile/random-forest job-exit-risk models, each calibrated to the same BRES-composition-adjusted manufacturing transition targets and converted into within-division displacement probabilities that preserve every division’s expected wage-bill loss. These are predictive baseline-transition benchmarks, not tariff-specific worker risks; their designs, distributions and fiscal sensitivity tables are in the Online Supplement (Sections S3–S4).

### 3.6 Universal Credit take-up after the shock

The take-up experiment is applied symmetrically across margins. After each shock, changed benefit units whose all-claim UC award moves from zero at baseline to positive post-shock

receive a claiming draw at the scenario take-up rate. This includes wage-cut households that cross into entitlement. Existing claimants, existing eligible non-claimants, unchanged units and post-shock-ineligible units retain their baseline flag, avoiding an unintended population-wide take-up reform. The grid therefore measures sensitivity to a common new-entitlement convention; it does not estimate actual claiming behaviour.

One modelling choice deserves its own subsection, because an earlier version of this paper got it wrong; the correction turns out to leave the unit-stress headline unchanged (Section 4.2), but that is a measured result, not a licence for the incorrect convention. PolicyEngine UK carries benefit take-up as a benefit-unit-level *stored input*, `would_claim_uc`, drawn once at dataset build time. Two properties of that draw matter. First, its target is a flat population-wide parameter—0.55 as a share of *all* benefit units, with reported UC recipients anchored to `TRUE` and the remainder filled to hit the target. It is not a take-up rate among the entitled, and it is therefore not comparable to the roughly 80 per cent take-up-among-entitled that the Resolution Foundation and the IFS assume; in the dataset used here the realised weighted flag rate is 0.547 over all benefit units and 0.541 among UC-eligible ones. Second, the flag is drawn on *pre-shock* circumstances. Nothing in a naive shock pipeline re-draws it, so a worker displaced by the shock carries into unemployment the claiming decision attributed to them while they were employed and, in the great majority of cases, not entitled to anything—measured take-up among the displaced post-shock was 46.9 per cent, below even the population flag rate. The baseline flag is thus not informative about the post-shock claiming behaviour of a newly entitled family: it encodes a decision taken in a state of the world the shock has destroyed.

The pipeline therefore re-draws `would_claim_uc` after each shock only for benefit units whose circumstances changed and whose all-claim UC award moves from zero at baseline to positive post-shock, at a scenario parameter `uc_takeup` with default 0.80. The re-draw uses an independent random stream, leaving worker assignment unchanged and preserving paired comparisons. This rule applies to displacement, inactivity, reallocation and wage cuts alike; existing eligible, post-shock-ineligible and unchanged units retain their baseline flag. Because the entire displacement-side cushion runs through UC, take-up is treated as a headline sensitivity rather than an exploratory one: Section 4.2 reports the primary displacement margin at new-entitlement take-up rates of 0.55, 0.80 and 1.00 over the full comparator draw count. The 0.80 default is a strong assumption for a population with no UC claiming history and unobserved capital (redundancy payments would gate many claims under the frozen capital thresholds); the grid brackets it rather than estimating it.

### 3.7 Microsimulation and outcomes

The Monte Carlo draws describe sensitivity to the artificial assignment of displacement across weighted FRS records. Their standard deviations are assignment dispersion, not sampling standard errors, confidence intervals, or uncertainty about the tariff response, survey design or model parameters. Numerical Monte Carlo error of a reported mean is the assignment SD divided by the square root of the number of draws. Parameter scenarios vary the demand calibration, wage-bill incidence, take-up and adjustment composition, but are not assigned probabilities and do not form confidence intervals. Survey sampling variance is approximated for the primary contrast by a household bootstrap of the FRS sample (Section 4.2); the research file carries no PSU or stratum identifiers, so the bootstrap treats households as independent and cannot reflect geographic clustering. Uncertainty in the trade-to-earnings bridge is not estimated. Reported comparisons other than the bootstrapped primary contrast are descriptive scenario contrasts.

Distributional cash-income changes use annual household disposable-income change divided by household size before person-weighted aggregation; aggregate disposable-income totals remain household-weighted totals.

The realisation step—an imposed status and earnings transition on survey households, then run through the full benefit rules—follows [Brewer and Tasseva \(2021\)](#), who apply it to the Covid-19 labour-market shock in UKMOD, and the EUROMOD nowcasting tradition of imposed labour-market transitions ([Navicke et al., 2013](#)). PolicyEngine UK computes baseline and shocked household disposable income (HBAI cash concept throughout), relative before-housing-costs poverty measured against a poverty line frozen at 60 per cent of the baseline equivalised median (so the reported change reflects households crossing a fixed threshold rather than mechanical movement of the line itself), absolute before-housing-costs poverty against a fixed baseline line, after-housing-costs poverty, the Gini coefficient, distributional breakdowns and the Exchequer effect. The *Exchequer effect* is defined as the change in the aggregate simulated government balance—all modelled tax and contribution revenue (including employer National Insurance contributions) minus all modelled benefit spending—between the baseline and shocked simulations. It is therefore broader than the employee-earnings loss  $\Delta E$  that enters the cushioning rate; Appendix A reconciles the two. On timing: trade exposure is built from 2024 HMRC customs data, the household input is FRS 2024–25, and the simulation year is 2026, with PolicyEngine UK’s standard uprating applied to survey monetary variables between the survey year and the simulation year. Primary submission scenarios use 50 balanced assignments; legacy direct and measured scenarios use 100 assignments, and explicitly exploratory diagnostics use five. Following [Ignatenko et al. \(2025\)](#), revenue recycling can flip aggregate employment responses; here fiscal policy is held fixed. Only direct taxes and benefits respond. Indirect taxes are not shocked, so reduced consumption-tax receipts are absent from the government balance.

## 4 Results

All results use FRS 2024–25 and period 2026. The bounds table uses 50 balanced assignments; legacy secondary scenarios use 100. Values after  $\pm$  are assignment standard deviations, not standard errors or confidence intervals. Numerical Monte Carlo SE is stored separately in each JSON artifact. Five-assignment diagnostics are exploratory.

### 4.1 Derived sector shocks

The deterministic OBR-style lower-anchor stress is £354 million annually. The unit-stress full-schedule amount is £886 million and the EPD unit stress is £693 million. These are scenario inputs generated by Equation (1), not estimated causal effects. The unit-stress sector rates are exactly half those of the former  $\varepsilon = 2$  high case and 2.5 times the OBR-style rates.

### 4.2 Cushioning

Table 1 reports the two polar bounds of the incidence surface: it crosses the external OBR-style and deliberately stronger unit anchors with one common-loss contrast and three duration-equivalent scales. At the 12-month unit stress, wage-cut cushioning exceeds displacement cushioning by 10.0 percentage points (assignment SD 4.7), while its Exchequer cost differs by £107 million. These are conditional statutory comparisons, not confidence intervals for tariff effects.

Table 1: Primary common-loss contrast by anchor and duration-equivalent annual stress, 50 assignments

| Anchor | Months | Margin       | Gross loss<br>(£m) | Exchequer<br>(£m) | Cushioning<br>(%) | Poverty<br>(pp) |
|--------|--------|--------------|--------------------|-------------------|-------------------|-----------------|
| OBR    | 6      | Displacement | 177 ± 3            | 73 ± 13           | 32.7 ± 7.7        | 0.009           |
| OBR    | 6      | Wage cut     | 177                | 96                | 43.7              | 0.000           |
| OBR    | 12     | Displacement | 354 ± 4            | 141 ± 21          | 32.0 ± 5.7        | 0.015           |
| OBR    | 12     | Wage cut     | 354                | 192               | 43.6              | 0.000           |
| UNIT   | 3      | Displacement | 222 ± 4            | 91 ± 15           | 32.0 ± 6.2        | 0.008           |
| UNIT   | 3      | Wage cut     | 221                | 120               | 43.6              | 0.000           |
| UNIT   | 6      | Displacement | 442 ± 6            | 190 ± 24          | 33.8 ± 5.4        | 0.016           |
| UNIT   | 6      | Wage cut     | 443                | 241               | 43.6              | 0.000           |
| UNIT   | 12     | Displacement | 885 ± 6            | 378 ± 37          | 34.0 ± 4.7        | 0.036           |
| UNIT   | 12     | Wage cut     | 886                | 484               | 43.9              | 0.000           |

*Notes:* OBR uses export-demand elasticity 0.4; unit uses 1.0. Values after ± are assignment SDs. Displacement uses balanced numerical integration; wage cuts are deterministic apart from the common UC claiming redraw. Three- and six-month rows scale the annual probability of a coherent full-year displaced state; they match expected worker-month loss but do not simulate partial-year unemployment spells (Appendix E bounds the annualisation bias). The OBR three-month cell is omitted: its displacement draw rests on 1.2 effective record contributions and is reported in Appendix C as sensitivity evidence only. Poverty is the change in the person-weighted fixed-line national BHC rate; because the exposed population is small and largely above the poverty line, near-zero national changes carry no information about losses among affected households.

The mixed grid below is the conceptual centre of the design; the bounds make its composition effect transparent.

*Estimator provenance.* Every headline number in the abstract and this section comes from the same 50-assignment paired-draw design: the wage-cut and displacement cushioning levels are draw means of the two unit 12-month scenarios, and the 10.0-point gap is the mean of the per-draw differences on paired seeds, so the displayed levels and gap are consistent up to rounding of the unrounded underlying estimates. The record-resampling contrast reported below (10.7 points) is a different estimator—the draw-averaged contrast over the 25 common assignment draws retained in the resampling cache—and is labelled as such wherever it appears.

### 4.3 Factorial decomposition of the gap

The two headline scenarios differ in two respects at once: the adjustment margin and the concentration and identity of the losers. A diffuse wage cut spreads the loss over all exposed workers; displacement concentrates it on roughly ten selected survey records. To separate the channels, a factorial middle cell—a *concentrated wage cut*—imposes the displacement draw’s exact worker-level losses (the same balanced draw on the same seeds, each drawn worker losing their entire annual earnings, with pension contributions and statutory pay scaling to zero exactly as for any in-work earnings change) while leaving every worker employed at baseline hours. Because employment is binary, a “diffuse displacement” cell is not definable, so the sequential path diffuse → concentrated → displaced is the unique monotone decomposition of the gap (and coincides with the Shapley allocation over the two factors).

At the unit 12-month stress over 50 paired assignments, the concentrated wage cut is cushioned at 34.1 per cent, almost exactly the displacement rate (34.0) and far below the diffuse

wage cut (43.9). The decomposition is one-sided: of the total  $10.0 \pm 4.7$ -point gap,  $9.9 \pm 4.7$  points arise from concentration and worker selection while employment state is held fixed, and only  $0.1 \pm 0.2$  points—about 1 per cent of the gap—from the employment-state change itself, holding the loss-bearing workers and their worker-level losses fixed. At the OBR-style anchor the pattern repeats (concentration 11.4 points, employment state 0.1 points). The channel split on common seeds locates the concentration step almost entirely in the income-tax schedule: diffuse cuts retain 16.1 points more marginal-rate income-tax relief than the concentrated cell, partially offset by £ for £ stronger Universal Credit support of the concentrated losses (-1.8 points in the concentrated cell’s favour). The employment-state step is nearly empty in every channel (UC 0.2 points; income tax and National Insurance zero by construction, since both cells zero the same earnings).

Two readings follow, and both matter. First, the paper’s earlier language of a “statutory asymmetry between adjustment margins” was imprecise: within the modelled annual system, the diffuse-wage-cut versus displacement gap is a *loss-concentration* effect—complete individual losses exhaust marginal-rate relief—not an effect of unemployment status, which Universal Credit’s income-based means test barely registers. Second, the near-zero employment-state step measures the modelled statutory rules, not the lived institutions of job loss: New Style JSA, monthly assessment periods, the five-week wait, claiming frictions and work-search conditionality all condition on employment status and all sit outside the annual model. The decomposition therefore bounds what a static statutory reading can attribute to the extensive margin, and it shifts the substantive weight of the headline contrast onto how concentrated the earnings losses are.

**Assignment-design and record-support audit.** At the unit 12-month stress, balanced and Bernoulli integration agree closely: cushioning is 34.0 versus 33.5 per cent and Exchequer cost is £378 versus £377 million. Balancing reduces gross-loss SD to 0.02 of the Bernoulli value and Exchequer SD to 0.18, but it cannot repair sparse survey support. The unit displacement loss has only 6.2 effective record contributions and its largest record supplies 28.7 per cent on average. At the OBR three-month scale these deteriorate to 1.2 and 91.9 per cent. The smallest displacement rows are consequently sensitivity evidence, not precise population estimates. Across leave-each-sector-out reruns for all 23 exposed divisions, the wage-minus-displacement cushioning difference remains positive, ranging from 7.6 to 11.3 percentage points. No single division creates or reverses the primary contrast.

**Record-resampling sensitivity.** Assignment SDs do not measure sensitivity to which households the FRS sampled, so the primary contrast is additionally scored with a household record-resampling exercise: 999 multinomial resamples of the 16,288 households, applied to per-household contributions from 25 common assignment draws of each unit 12-month margin, so that resampling variation is measured around the draw-averaged statistic. On that estimator the wage-minus-displacement cushioning difference is 10.7 percentage points with a resampling standard error of 1.2 points and a 2.5th–97.5th percentile range of 9.1 to 13.9 points. This is deliberately not presented as a significance test: the two counterfactuals are imposed scenarios, so their contrast is partly generated by construction, and the resampling captures record composition only—not the FRS complex survey design (no PSU or stratum identifiers exist in the research file), not tariff-calibration, incidence, concordance or model-parameter uncertainty, and not behavioural or claiming uncertainty, all of which remain scenario dimensions. The range should

be read as robustness of the contrast to record composition, nothing stronger. Margin-level resampling SEs are 1.2 points (displacement cushioning), 0.9 points (wage-cut cushioning), and £43 million and £37 million on the respective Exchequer costs. One convention should be noted: a resample can zero a displacement draw’s gross loss, leaving that draw’s cushioning ratio undefined; replicates average over the defined draws, and near-zero-gross resamples with unstable ratios are retained.

**Leave-one-record-out.** Because the displacement side of the contrast rests on few effective records, each of the 194 FRS households that contributes a positive weighted gross loss to any of the 25 cached displacement draws is removed in turn—from both margins, since the same survey record underlies both counterfactuals—and the draw-averaged contrast is recomputed exactly from stored per-household contributions. The contrast remains positive in every case, ranging from 10.3 to 11.3 percentage points around the 10.7-point resampling-cache value; the most influential single record moves it by 0.6 points. No individual FRS household creates the headline gap, although the sparse support continues to rule out subgroup, regional and constituency readings of the displacement scenarios.

**Pension-contribution channel.** The HBAI disposable-income concept treats employee pension and salary-sacrifice contributions as an outflow, so a hypothesis worth testing is that part of the headline gap is reduced retirement saving rather than tax-benefit insurance. It is not. Recomputing both margins under a pension-gross income concept (disposable income plus employee pension and salary-sacrifice contributions, unit 12-month stress, the primary balanced design over 50 paired assignments) lowers wage-cut cushioning from 43.9 to 39.6 per cent and displacement cushioning from 34.0 to 29.1 per cent—the pension outflow falls by close to the same share of the gross loss on both margins (4.4 and 4.9 per cent respectively; wage cuts scale contributions, displacement zeroes them)—so the wage-minus-displacement gap moves only from 10.0 to 10.5 percentage points. The headline contrast is a tax-benefit schedule effect, not a pension-accounting artefact. What the pension channel does affect is the *level* of measured cushioning on both margins, and the implied loss of deferred retirement consumption is not captured by the cushioning rate.

**Mixed and transition paths.** The illustrative interior calibration is the transition-equivalent path: 70 per cent of each sector’s declared loss is assigned to workers whose annual earnings and hours embody a three-month re-employment lag before entering a services job at the FRS 28.3 per cent earnings penalty, with survivor cuts rescaled so the expected sector loss remains common. It yields  $41.4 \pm 2.4$  per cent cushioning and an Exchequer cost of  $£461 \pm 135$  million—between the bounds, as the composition logic implies. The surrounding response surface (Figure 1) includes a churn-dominated 85/15 displacement-survivor split ( $35.9 \pm 4.3$  per cent cushioning,  $£406 \pm 210$  million) and the lower 14 per cent penalty sensitivity reported in the appendix. The 70/30 share, lag and penalty are assumptions bounded by external evidence, not estimated tariff responses.

#### 4.4 External magnitude and outturn checks

The frozen HMRC customs build totals £55.6 billion of 2024 US-bound goods exports before exclusions and maps £52.5 billion to producing divisions. This is reasonably close to the ONS balance-of-payments headline of £59.3 billion, given known conceptual and coverage differences

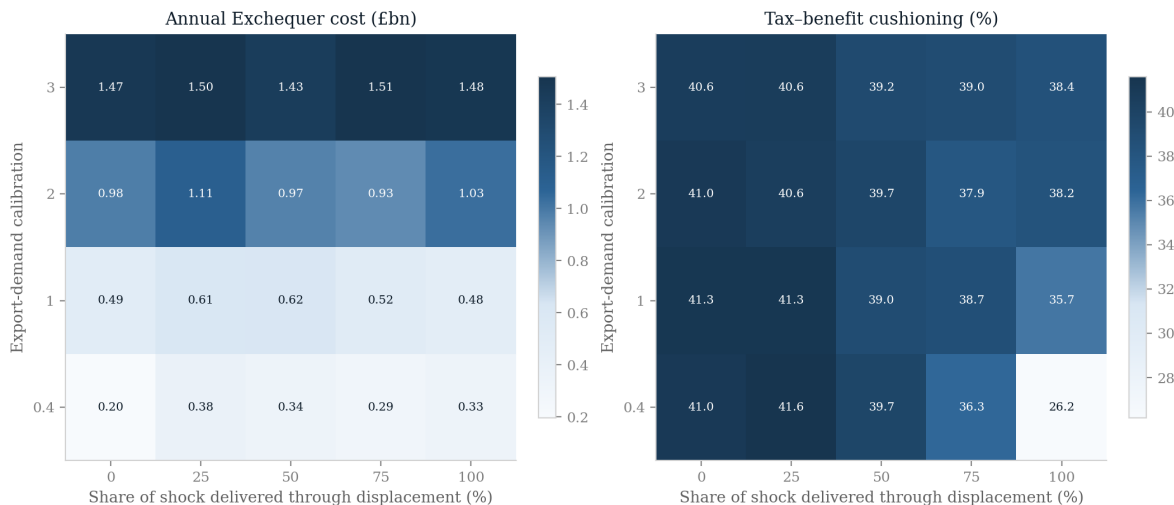


Figure 1: Factorial scenario surface: export-demand calibration by the share of expected sector loss delivered through displacement rather than survivor wage cuts. Cell values are means over five common assignments.

(ONS, 2025). The reference full-schedule calibration predicts a 12.57 per cent aggregate fall across mapped divisions. The observed May 2025–February 2026 customs window falls 12.71 per cent against its two-year same-month baseline. This near equality is not a validation result: the reference calibration was not estimated from that window, and the outturn combines tariff effects with prices, exchange rates, anticipation, composition, rerouting and other demand shocks.

Sector agreement is much weaker. The stored validation artifact reports a rank correlation of 0.00 between the full-schedule predicted and realised falls among the ten largest exposed divisions, and a correlation of 0.58 between full-schedule and observed downside shock rates across all mapped divisions. Machinery and motor vehicles decline broadly in the expected direction, while aerospace, electrical equipment and basic metals do not. The aggregate match can therefore be coincidental and must not be used to validate Equation (1). As a separate policy benchmark, the OBR’s March 2025 scenario uses a demand elasticity of 0.4, implying an 8 per cent export-demand fall after a 20 percentage-point tariff and a direct export-demand effect just under 0.2 per cent of GDP (OBR, 2025). The paper’s  $\varepsilon = 1$  reference is 2.5 times that elasticity and additionally assumes full wage-bill incidence; it is a deliberately strong stress, not a central forecast.

**Product–destination panel.** The monthly HMRC panel (Online Supplement, Sections S1–S2) supports a post-policy decline concentrated in large US export products—the capped value-weighted specification estimates a -31.9 per cent change in US-bound exports relative to the same products sent to Canada, Japan and Australia—but the tariff-intensity ordering fails: broad steel and auto chapters fall -20.0 per cent against -32.9 per cent for other products. As set out in the introduction, this failed falsification tests the bridge, not the estimand: the cushioning comparison conditions on a declared loss that is identical across margins by construction, and the leave-one-division-out audit (all 23 divisions, contrast range 7.6–11.3 points) together with the three LFS-shaped selection models confirm that sector composition does not create or reverse it. What the failure rules out is any reading of Equation (1) as an estimated first stage.

**Longitudinal worker benchmark (summary).** A five-quarter longitudinal LFS benchmark (Study 9490) calibrates cell, earnings-band and quantile-random-forest job-exit-risk models to BRES-adjusted manufacturing targets and propagates each through the fiscal model while preserving every division’s expected wage-bill loss. Across all three LFS-shaped risk models, mean cushioning spans 36.0–37.4 per cent against 36.8 under uniform assignment, a spread smaller than the assignment SD of any single specification; person-level agreement between the models is weak, so they quantify selection-model sensitivity rather than identify tariff-specific worker risk. Full designs, distributions and tables are in the Online Supplement (Sections S3–S4).

The realised displacement output is also consistent with the allocation design but too sparse for precise empirical incidence: the expected 9.7 selected FRS records per assignment produce 17.2 thousand represented displaced workers and  $\pounds882 \pm 418$  million of gross loss. The gross-loss SD is nearly half its mean, and Exchequer-cost SD exceeds half its mean. The 100-assignment average is numerically more stable than an individual assignment, but it remains conditional on the observed 1,018 exposed records and the uniform within-division selection rule.

**Take-up as a headline sensitivity.** Because the displacement-side cushion runs through UC, take-up is scored at the full comparator draw count (50 Bernoulli assignments) rather than as an exploratory grid, with the new-entitlement claiming rate at 0.55, 0.80 and 1.00 and, additionally, the stale pre-shock claiming flag restored as a fourth convention. The measured answer is that none of these moves the unit-stress headline: displacement cushioning is 33.0 per cent under every convention, with differences below 0.1 percentage points—within the numerical precision of the aggregation. The reason is institutional and was not obvious *ex ante*: most displaced households are *already* entitled to UC at baseline through the in-work means test, so under the maintained rule they retain their stored claiming state, and the newly entitled population that the take-up parameter governs carries little survey weight. The residual convention that this grid cannot test is the retention of baseline claiming states for already-entitled non-claimants—about half of entitled benefit units in the input data do not claim—and the unmodelled UC capital test (redundancy payments against thresholds frozen since 2006), both of which would move measured cushioning down, not up (Section 6). Wage-cut results are take-up-insensitive because few wage-cut households change UC entitlement.

## 4.5 Mechanism

The regenerated decomposition and gate audit are five-assignment diagnostics, but they clarify why the two margins produce different fiscal outcomes. A wage cut keeps the worker inside the income-tax and National Insurance schedule, so relief is concentrated in lower tax and contribution payments. Displacement removes earnings altogether and shifts the response towards Universal Credit and other means-tested support. The latter channel is nonlinear: entitlement depends on household composition, existing income, rent, capital, and claiming behaviour rather than on the individual earnings loss alone. The plain FRS input contains no usable household-capital measure, so the model cannot establish the empirical importance of UC’s capital test. New-style JSA is not activated by the imposed transition. The gate audit should therefore be read as a decomposition of the implemented rules, not as a complete account of the UK out-of-work system.

## 4.6 Distributional incidence

Under full displacement, relative BHC poverty rises by  $0.032 \pm 0.022$  percentage points. Under wage cuts it is unchanged at displayed precision. The small national change does not imply that exposed households are protected: the affected population is limited, and many exposed workers begin above the poverty threshold. The decile profile shows a broad reduction in disposable income, with the largest losses concentrated towards the upper end of the distribution where exposed earnings are concentrated. This pattern is consistent with a progressive tax–benefit response that cushions lower-income households more strongly than the gross earnings measure. These figures are distributional diagnostics rather than population-wide poverty forecasts. Figure 2 shows the decile profile.

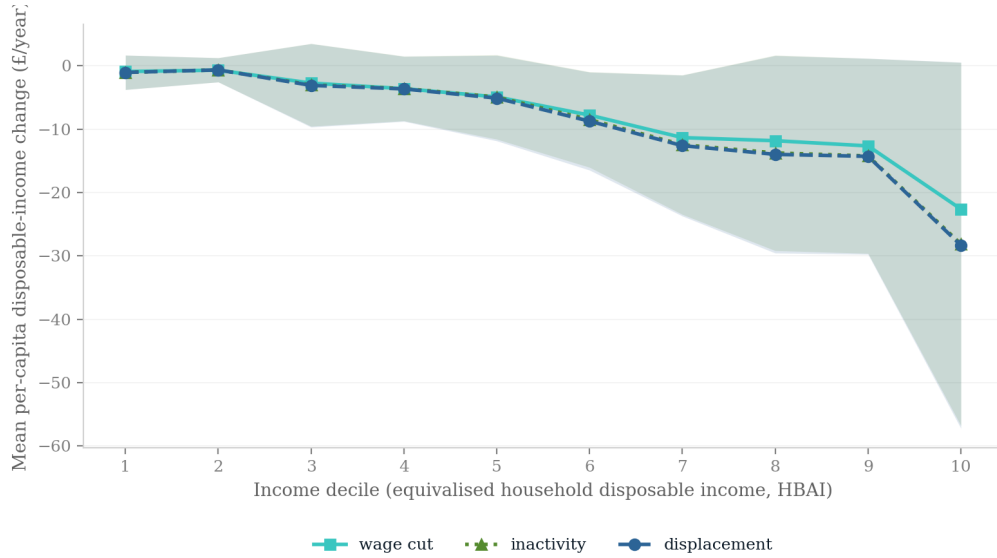


Figure 2: Per-capita disposable-income change by baseline income decile in the full-schedule scenarios. Shading denotes assignment dispersion.

A constituency-level surface was produced in earlier drafts but is not reported: it rests on imputed industry assignments rather than observed local industry structure, and a map invites exactly the seat-ranking reading it cannot support. The Online Supplement documents the machinery for future use with observed local industry data.

## 4.7 Fiscal incidence

Annual Exchequer costs are reported in Table 1. The measure is the aggregate simulated government-balance change and is broader than the household cushioning identity. The cost combines lower income-tax and National Insurance receipts with additional benefit payments. Employer National Insurance contributions and the tax consequences of pension and salary-sacrifice changes respond to the shock but sit outside employee disposable income, which explains why the Exchequer cost does not equal the household-side cushioning amount. Displacement also produces more assignment dispersion because a small number of high-weight records can determine whether earnings are lost entirely. Appendix A provides the formal reconciliation and draw distributions. Figure 3 shows the assignment distribution.



Figure 3: Annual Exchequer cost across regenerated assignments. The spread is assignment dispersion from the Monte Carlo allocation of displacement, not sampling uncertainty.

## 4.8 EPD counterfactual

On paired seeds, full minus EPD displacement differs by  $3.7 \pm 2.8$  thousand workers and  $\text{£}90 \pm 98$  million of Exchequer cost, driven primarily by the lower auto tariff and the pharmaceutical exemption. It prices a change in the declared sector shock with the household model and draws held fixed, not the deal’s causal effect; the paired comparison and its caveats are in the Online Supplement (Section S5).

# 5 Policy

## 5.1 Economic Prosperity Deal

Within the maintained direct-channel scenarios, the paired full-minus-EPD displacement contrast is  $3.7 \pm 2.8$  thousand workers,  $\text{£}204 \pm 192$  million of gross earnings and  $\text{£}90 \pm 98$  million of annual Exchequer cost. These SDs describe assignment dispersion and the contrasts are not causal estimates of the deal. Autos drive much of the difference; the scenario applies the in-quota rate to fixed exposure without modelling quota allocation or future above-quota trade.

## 5.2 Pharmaceutical channel

The EPD scenario sets the pharmaceutical employment shock to zero, but this does not make the exemption free. Its VPAG and NHS pricing concessions sit outside the labour-income microsimulation. A complete policy appraisal would cost those health-budget and price effects alongside the direct tax–benefit scenario.

### 5.3 Cushioning policy

The simulations support a limited conclusion: extensive-margin protection depends more on means-tested eligibility and claiming than the income-tax response to an intensive earnings loss. The full-draw take-up grid shows the claiming-rate assumption itself is not the binding parameter—most displaced households are already UC-entitled in work—which shifts the policy question towards the gates the model cannot exercise: the capital test and take-up among long-standing eligible non-claimants. Automatic enrolment after redundancy, contribution-based unemployment insurance and wage insurance are therefore candidate counterfactuals for future analysis, not policy recommendations established by the present results. Their costs and incidence require dedicated simulations and financing assumptions.

The repository also supplies reusable interfaces for the next policy exercises. Observable household and worker characteristics can tilt shock probabilities while preserving each sector’s aggregate earnings loss, and a separate price-channel interface converts expenditure-weighted price changes into real disposable income. Neither interface supplies causal coefficients: the former requires externally estimated heterogeneity and the latter requires household expenditure data. They are extensions to populate and validate, not additional headline estimates in this paper.

## 6 Discussion and conclusion

The central finding is that the UK tax–benefit system responds differently when a common aggregate earnings loss is distributed differently across workers and labour-market states. Across two loss anchors and three duration-equivalent scales, the wage-cut and displacement bounds differ by 9.8–12.8 percentage points; at the unit twelve-month scale the difference is 10.0 points and the wage-cut Exchequer cost is £107 million higher. The factorial middle cell splits that difference into 9.9 points from concentrating the loss on selected workers and only 0.1 points from the employment-state change itself; the record-resampling exercise places the total contrast at 10.7 points (sensitivity range 9.1–13.9), and it survives the removal of any single loss-contributing record (10.3–11.3 points). The mechanism is a loss-concentration effect inside the statutory schedule: diffuse cuts are relieved at marginal income-tax rates, complete individual losses exhaust that relief and land on means-tested UC, and the unemployment state itself adds almost nothing within the modelled annual rules—a bound on static statutory readings of the extensive margin, since the status-contingent institutions (New Style JSA, monthly assessment, waiting periods, conditionality) are unmodelled. On the UC side four objects must be kept distinct: statutory entitlement as represented by PolicyEngine; entitlement conditional on the incomplete household information in the FRS (no usable liquid-capital measure, so the capital gates are not exercised); modelled take-up, which is a scenario convention; and actual receipt following real job loss, which none of the above observes. The simulation evidence speaks to the first three. Two sensitivity results sharpen this reading. Under a pension-gross income concept the cushioning *levels* fall by roughly four percentage points on both margins but the contrast is essentially unchanged, so the gap is a statutory asymmetry, not an artefact of the HBAI treatment of pension saving. And the take-up grid shows the gap is not an artefact of the claiming convention either: varying the new-entitlement rate from 0.55 to 1.00, or restoring the stale pre-shock flag, leaves the unit-stress displacement cushioning unchanged to within 0.1 points, because most displaced households are already UC-entitled in work. The contrast is pre-specified and conditional: it prices a statutory asymmetry, not the tariffs’ total causal incidence.

The annualisation of unemployment spells is conservative for it: the parameter-based bounding in Appendix E shows that correct monthly UC and partial-year PAYE accounting would raise displacement cushioning at sub-annual durations and leave the 12-month contrast exact.

Balanced integration makes the common aggregate loss comparable assignment by assignment and closely reproduces the Bernoulli mean at unit scale. It does not overcome sparse information: the unit displacement loss has only 6.2 effective record contributions, and support is weaker at the smallest OBR-duration cells. Assignment SDs describe scenario-allocation sensitivity; record-composition sensitivity is scored separately by the household resampling exercise and the leave-one-record-out audit, neither of which is a design-based confidence interval, and no significance claim is attached to any interval. The sign also survives all 23 leave-one-division-out reruns, with a cushioning difference of 7.6–11.3 points. This establishes sector robustness within the imposed bridge, not causal robustness of that bridge.

Calibrated longitudinal-LFS allocation sensitivities leave the central extensive-margin conclusion broadly unchanged: cushioning spans 36.0–37.4 per cent across cell, earnings-band and regularised-QRF risk shapes, compared with 36.8 per cent under uniform assignment. This robustness is conditional and limited. The imputation models agree weakly person by person, and the available LFS manufacturing target has only 177 donors; stability of an aggregate fiscal ratio is not evidence that tariff-specific worker selection has been identified.

National poverty changes remain near zero because the exposed population is small and often above the poverty line. They are not precise population estimates and do not imply that affected households' losses are small. EPD, inactivity, reallocation and demographic results are secondary or supplementary rather than competing headline estimands.

The common-loss transition-equivalent path makes the interior of the surface concrete: some workers receive an annualized re-employment penalty, some re-enter services at lower earnings, and survivors absorb the residual loss. Under the illustrative 70/30 transition allocation, the simulated cushioning is 41.4 per cent and the Exchequer cost is £461 million. These are conditional results: the annualized lag is encoded in earnings and hours rather than a monthly unemployment-benefit spell; the transition share, three-month lag and 28.3 per cent services penalty are scenario inputs, and the latter is a cross-sectional FRS destination gap that also captures hours and part-time composition. The response surface, including 50/50 and 85/15 allocations and the 14 per cent penalty, is therefore more informative than any single preferred point.

The principal limitation is that the analysis begins after the production-side transmission mechanism. It does not estimate how tariffs affect prices, quantities, output, value added, imported inputs, productivity, investment, profits, or labour demand. Instead, it maps a declared sector earnings loss directly onto households and evaluates the statutory response. The resulting estimates therefore identify fiscal incidence conditional on the imposed loss and adjustment path. They do not identify the economy-wide effect of the trade policy. Worker selection is independent within broad SIC divisions; the longitudinal LFS models provide only weakly concordant predictive rankings; within-year benefit timing, New Style JSA, household labour supply, indirect taxes, exchange rates, retaliation, and general equilibrium remain outside the central specification. Constituency machinery in the Online Supplement is illustrative because local industry structure is not observed at the required level.

The framework nevertheless provides a useful platform for subsequent work. The next empirical step is to combine product- or firm-level evidence on tariff exposure, prices, quantities, output, and worker outcomes with the household microsimulation. Such evidence would re-

place the imposed earnings bridge with an estimated transmission mechanism while retaining the adjustment margin and fiscal-incidence decomposition developed here.

The main conclusion is that, conditional on a common earnings loss, the labour-market margin materially changes the distribution of tax relief, benefit support, poverty effects, and public-finance costs. Wage-cut and displacement specifications are transparent bounds; the mixed and transition-equivalent paths show how intermediate adjustment patterns alter those outcomes. The results therefore speak directly to the fiscal consequences of labour-market adjustment, while leaving the size and composition of the underlying trade shock as an empirical question for future research.

## References

- Office for National Statistics (2025). *Labour Force Survey Five-Quarter Longitudinal Dataset, July 2024–September 2025*. UK Data Service, Study Number 9490. doi:10.5255/UKDA-SN-9490-1.
- HM Revenue and Customs (2026). *UK overseas trade in goods statistics: detailed commodity by country data*. HMRC Online Trade Statistics API, monthly observations through May 2026. <https://www.uktradeinfo.com/>.
- ABPI (2025). UK–US agreement on pharmaceutical tariffs and pricing. Association of the British Pharmaceutical Industry, London, December 2025.
- Acemoglu, D., Akcigit, U. and Kerr, W. (2016). Networks and the macroeconomy: An empirical exploration. *NBER Macroeconomics Annual*, 30(1), 273–335.
- Acosta, M. and Cox, L. (2024). The regressive nature of the US tariff code: Origins and implications. Working paper, February 2024 (conditionally accepted, *The Quarterly Journal of Economics*).
- Adão, R., Arkolakis, C. and Esposito, F. (2023). General equilibrium effects in space: Theory and measurement. NBER Working Paper No. 25544, National Bureau of Economic Research, Cambridge, MA.
- Amiti, M., Redding, S. J. and Weinstein, D. E. (2019). The impact of the 2018 tariffs on prices and welfare. *Journal of Economic Perspectives*, 33(4), 187–210.
- Aragón, F. M., Rud, J. P. and Toews, G. (2018). Resource shocks, employment, and gender: Evidence from the collapse of the UK coal industry. *Labour Economics*, 52, 54–67.
- Artuc, E., Porto, G. and Rijkers, B. (2021). Household impacts of tariffs: Data and results from agricultural trade protection. *The World Bank Economic Review*, 35(3), 563–585.
- Auerbach, A. J. and Feenberg, D. (2000). The significance of federal taxes as automatic stabilizers. *Journal of Economic Perspectives*, 14(3), 37–56.
- Atkin, D., Bernadac, V., Donaldson, D., Garg, T. and Huneus, F. (2025). The incidence of distortions. Working paper, December 2025 draft.
- Auray, S., Devereux, M. B. and Eyquem, A. (2020). Trade wars, currency wars. NBER Working Paper No. 27460, National Bureau of Economic Research, Cambridge, MA.
- Autor, D. H., Dorn, D. and Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, 103(6), 2121–2168.
- Autor, D. H., Dorn, D., Hanson, G. H. and Song, J. (2014). Trade adjustment: Worker-level evidence. *The Quarterly Journal of Economics*, 129(4), 1799–1860.
- Autor, D. H., Dorn, D. and Hanson, G. H. (2021). On the persistence of the China shock. *Brookings Papers on Economic Activity*, Fall 2021, 381–447.

- Bank of England (2025). *Monetary Policy Report*, May, August and November 2025 issues. Bank of England, London.
- Barrot, J.-N. and Sauvagnat, J. (2016). Input specificity and the propagation of idiosyncratic shocks in production networks. *The Quarterly Journal of Economics*, 131(3), 1543–1592.
- Beatty, C. and Fothergill, S. (2017). The impact on welfare and public finances of job loss in industrial Britain. *Regional Studies, Regional Science*, 4(1), 161–180.
- Beatty, C., Fothergill, S. and Powell, R. (2007). Twenty years on: Has the economy of the UK coalfields recovered? *Environment and Planning A*, 39(7), 1654–1675.
- Beatty, C. and Fothergill, S. (2023). The persistence of hidden unemployment among incapacity claimants in large parts of Britain. *Local Economy*, doi:10.1177/02690942231184815.
- Blanchard, E. J., Bown, C. P. and Chor, D. (2019). Did Trump’s trade war impact the 2018 election? NBER Working Paper No. 26434, National Bureau of Economic Research, Cambridge, MA.
- Bloom, N., Bunn, P., Chen, S., Mizen, P., Smietanka, P. and Thwaites, G. (2019). The impact of Brexit on UK firms. NBER Working Paper No. 26218, National Bureau of Economic Research, Cambridge, MA.
- Borusyak, K. and Jaravel, X. (2021). The distributional effects of trade: Theory and evidence from the United States. NBER Working Paper No. 28957, National Bureau of Economic Research, Cambridge, MA.
- Bourguignon, F., Bussolo, M. and Pereira da Silva, L. A. (eds.) (2008). *The Impact of Macroeconomic Policies on Poverty and Income Distribution: Macro-Micro Evaluation Techniques and Tools*. Palgrave Macmillan and World Bank, Washington, DC.
- Brewer, M. and Tasseva, I. V. (2021). Did the UK policy response to Covid-19 protect household incomes? *The Journal of Economic Inequality*, 19(3), 433–458.
- Breinlich, H., Leromain, E., Novy, D. and Sampson, T. (2022). The Brexit vote, inflation and UK living standards. *International Economic Review*, 63(1), 63–93.
- The Budget Lab (2025). Where we stand: The fiscal, economic, and distributional effects of all U.S. tariffs enacted in 2025. The Budget Lab at Yale, New Haven, CT.
- Caliendo, L., Dvorkin, M. and Parro, F. (2019). Trade and labor market dynamics: General equilibrium analysis of the China trade shock. *Econometrica*, 87(3), 741–835.
- Card, D., Cardoso, A. R., Heining, J. and Kline, P. (2018). Firms and labor market inequality: Evidence and some theory. *Journal of Labor Economics*, 36(S1), S13–S70.
- Cavallo, A., Gopinath, G., Neiman, B. and Tang, J. (2021). Tariff pass-through at the border and at the store: Evidence from US trade policy. *American Economic Review: Insights*, 3(1), 19–34.
- Cavallo, A., Llamas, P. and Vazquez, F. (2025). Tracking the short-run price impact of U.S. tariffs. NBER Working Paper No. 34496, National Bureau of Economic Research, Cambridge, MA.

- Centre for Economic Performance (2025). UK firms and the 2025 US tariffs: Evidence from the Decision Maker Panel. VoxEU column, Centre for Economic Policy Research, and CEP “in brief” 11639, Centre for Economic Performance, LSE, London. <https://cepr.org/voxeu/columns/impact-2025-us-tariff-announcements-uk-firms>.
- Centre for Cities (2025). The uneven storm: How US tariffs hit UK cities differently. Centre for Cities, London.
- City-REDI (2025). The regional impact of US automotive tariffs on the UK economy. City-REDI, University of Birmingham.
- Corsetti, G., Lloyd, S. and Ostry, D. (2025). Trading blows: The exchange-rate response to tariffs and retaliations. Staff Working Paper No. 1,139, Bank of England, London.
- Dauth, W., Findeisen, S. and Suedekum, J. (2021). Adjusting to globalization in Germany. *Journal of Labor Economics*, 39(1), 263–302.
- De Lyon, J. and Pessoa, J. P. (2021). Worker and firm responses to trade shocks: The UK–China case. *European Economic Review*, 133, 103678.
- Department for International Trade (2021). *Trade Modelling Review Expert Panel: Report and Recommendations*. Department for International Trade, London.
- den Besten, T. and Känzig, D. R. (2025). The macroeconomic effects of tariffs: Evidence from U.S. historical data. NBER Working Paper No. 34852, National Bureau of Economic Research, Cambridge, MA.
- Dhingra, S. (2025). Tariffed with the same brush. Speech, Bank of England, October 2025.
- Dhingra, S., Ottaviano, G., Sampson, T. and Van Reenen, J. (2017). The costs and benefits of leaving the EU: Trade effects. *Economic Policy*, 32(92), 651–705.
- Dix-Carneiro, R. (2014). Trade liberalization and labor market dynamics. *Econometrica*, 82(3), 825–885.
- Dix-Carneiro, R. and Kovak, B. K. (2017). Trade liberalization and regional dynamics. *American Economic Review*, 107(10), 2908–2946.
- Dolls, M., Fuest, C. and Peichl, A. (2012). Automatic stabilizers and economic crisis: US vs. Europe. *Journal of Public Economics*, 96(3–4), 279–294.
- Dorn, D. and Levell, P. (2024). Labour market impacts of the China shock: Why the tide of globalisation did not lift all boats. *Labour Economics*, 91, 102629.
- Department for Work and Pensions (2024). Income-related benefits: Estimates of take-up. Official statistics series, DWP, London.
- Fajgelbaum, P. D. and Khandelwal, A. K. (2016). Measuring the unequal gains from trade. *The Quarterly Journal of Economics*, 131(3), 1113–1180.
- Fajgelbaum, P. D. and Khandelwal, A. K. (2022). The economic impacts of the US–China trade war. *Annual Review of Economics*, 14, 205–228.

- Fajgelbaum, P. D. and Khandelwal, A. K. (2026). Tariffs in 2025: Short-run impacts on the U.S. economy. *Brookings Papers on Economic Activity*, Spring 2026 conference draft.
- Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J. and Khandelwal, A. K. (2020). The return to protectionism. *The Quarterly Journal of Economics*, 135(1), 1–55.
- Fetzer, T. (2019). Did austerity cause Brexit? *American Economic Review*, 109(11), 3849–3886.
- Fetzer, T. and Wang, S. (2024). Measuring the regional economic cost of Brexit: Evidence up to 2019. CAGE Working Paper No. 1280, University of Warwick.
- Flaaen, A. and Pierce, J. R. (2019). Disentangling the effects of the 2018–2019 tariffs on a globally connected U.S. manufacturing sector. Finance and Economics Discussion Series 2019-086, Board of Governors of the Federal Reserve System, Washington, DC.
- Galle, S., Rodríguez-Clare, A. and Yi, M. (2023). Slicing the pie: Quantifying the aggregate and distributional effects of trade. *The Review of Economic Studies*, 90(1), 331–375.
- Garin, A. and Silvério, F. (2024). How responsive are wages to firm-specific changes in labour demand? Evidence from idiosyncratic export demand shocks. *The Review of Economic Studies*, 91(3), 1671–1710.
- Gathmann, C., Helm, I. and Schönberg, U. (2020). Spillover effects of mass layoffs. *Journal of the European Economic Association*, 18(1), 427–468.
- Hummels, D., Jørgensen, R., Munch, J. and Xiang, C. (2014). The wage effects of offshoring: Evidence from Danish matched worker–firm data. *American Economic Review*, 104(6), 1597–1629.
- Hyman, B., Kovak, B. K. and Leive, A. (2024). Wage insurance for displaced workers. NBER Working Paper No. 32464, National Bureau of Economic Research, Cambridge, MA.
- Institute for Fiscal Studies (2024). Health-related benefit claims post-pandemic: UK trends and global context. IFS Report, Institute for Fiscal Studies, London, September 2024.
- International Monetary Fund (2025). *World Economic Outlook: Global Economy in Flux, Prospects Remain Dim*. IMF, Washington, DC, October 2025.
- Ignatenko, A., Lashkaripour, A., Macedoni, L. and Simonovska, I. (2025). Making America great again? The economic impacts of Liberation Day tariffs. *Journal of International Economics*, 157, 104138.
- Irastorza-Fadrique, A., Levell, P. and Parey, M. (2025). Household responses to trade shocks. *Journal of International Economics*, forthcoming (IFS Working Paper 24/52).
- Jeanne, O. and Son, J. (2024). To what extent are tariffs offset by exchange rates? *Journal of International Money and Finance*, 142.
- Kalemli-Özcan, Ş., Soylu, C. and Yildirim, M. A. (2026). Global trade, tariff uncertainty and the U.S. dollar. NBER Working Paper No. 34728, National Bureau of Economic Research, Cambridge, MA.

- Kim, R. and Vogel, J. (2021). Trade shocks and labor market adjustment. *American Economic Review: Insights*, 3(1), 115–130.
- Kline, P., Petkova, N., Williams, H. and Zidar, O. (2019). Who profits from patents? Rent-sharing at innovative firms. *The Quarterly Journal of Economics*, 134(3), 1343–1404.
- Levell, P., O’Connell, M. and Smith, K. (2017). The exposure of households’ food spending to tariff changes and exchange rate movements. IFS Briefing Note BN213, Institute for Fiscal Studies, London.
- Luu, N., Woloszko, N., Causa, O., Arriola, C., van Tongeren, F. and Johansson, Å. (2020). Mapping trade to household budget survey: A conversion framework for assessing the distributional impact of trade policies. OECD Trade Policy Papers No. 244, OECD Publishing, Paris.
- McKay, A. and Reis, R. (2016). The role of automatic stabilizers in the U.S. business cycle. *Econometrica*, 84(1), 141–194.
- Michelen, G., Ernst, C. and Bertin, P. (2025). Tariffs and labor markets: The employment impact of the recent trade conflict. A multiregional input–output analysis. International Labour Organization / arXiv:2512.11578, December 2025.
- Moretti, E. (2010). Local multipliers. *American Economic Review: Papers & Proceedings*, 100(2), 373–377.
- Navicke, J., Rastrigina, O. and Sutherland, H. (2013). Nowcasting indicators of poverty risk in the European Union: A microsimulation approach. EUROMOD Working Paper No. EM11/13, ISER, University of Essex.
- NIESR (2024). Implications of higher US tariffs for the UK economy. National Institute of Economic and Social Research, London, December 2024.
- Office for Budget Responsibility (2025). *Economic and Fiscal Outlook*, March 2025 (tariff scenario box). OBR, London.
- Office for National Statistics (2025). UK trade with the United States: 2024. ONS, Newport, 25 April. <https://www.ons.gov.uk/economy/nationalaccounts/balanceofpayments/articles/uktradewiththeunitedstates/2024>.
- Peichl, A. (2016). Linking microsimulation and CGE models. *International Journal of Microsimulation*, 9(1), 167–174.
- Resolution Foundation (2025a). Trump tariff turmoil. Resolution Foundation, London, April 2025.
- Resolution Foundation (2025b). Saving penalties: Reforming the capital rules in Universal Credit. Resolution Foundation, London, April 2025.
- Roberts, J. and Taylor, K. (2022). New evidence on disability benefit claims in Britain: The role of health and the local labour market. *Economica*, 89(353), 131–160.

- Society of Motor Manufacturers and Traders (2025). Written submission to the House of Commons Business and Trade Committee inquiry on UK–US trade (reference UST0041). UK Parliament written evidence, <https://committees.parliament.uk/writtenevidence/167352/html/>.
- Rodríguez-Clare, A., Ulate, M. and Vasquez, J. P. (2025). The 2025 trade war: Dynamic impacts across U.S. states and the global economy. NBER Working Paper No. 33792, National Bureau of Economic Research, Cambridge, MA.
- Rud, J. P., Simmons, M., Toews, G. and Aragón, F. (2022). Job displacement costs of phasing out coal. IZA Discussion Paper No. 15581.
- Sutherland, H. and Figari, F. (2013). EUROMOD: The European Union tax–benefit microsimulation model. *International Journal of Microsimulation*, 6(1), 4–26.
- Tax Policy Center (2025). Description of the Tax Policy Center’s tariff models. Urban–Brookings Tax Policy Center, Washington, DC.
- Traiberman, S. (2019). Occupations and import competition: Evidence from Denmark. *American Economic Review*, 109(12), 4260–4301.
- Waugh, M. E. (2019). The consumption response to trade shocks: Evidence from the US–China trade war. NBER Working Paper No. 26353, National Bureau of Economic Research, Cambridge, MA.
- World Trade Organization (2025). *Global Trade Outlook and Statistics: Update October 2025*. WTO, Geneva.

## A Supporting Appendix

US exports use HMRC uktradeinfo OTS/RTS data for 2024, mapped from SITC to SIC 2007 divisions and divided by the latest non-suppressed ONS Annual Business Survey turnover. Of a £55,600 million customs-basis total, £52,500 million maps to divisions after documented exclusions, chiefly non-monetary gold and precious metals. The measured window uses May 2025–February 2026 against the mean of the same calendar months one and two years earlier. Its 12.7 per cent aggregate fall is descriptive, not causal.

## B Shock mechanics

Each employee in exposed division  $j$  is independently selected with probability  $s_j$ . Weighted displaced counts and earnings therefore meet targets in expectation rather than exactly. Selected workers have employment income, hours, pension contributions and statutory pay zeroed. Wage cuts remove  $s_j$  of each exposed worker’s earnings deterministically. Reallocation uses the same selected workers, assigns a services destination, and applies the calibrated earnings penalty.

UC claiming is redrawn after every margin for benefit units whose circumstances changed and that are entitled post-shock. The LCWRA override sets the benefit-unit element to the maximum of its stored baseline value and one statutory annual element, preventing duplicate payment.

## C Monte Carlo and measures

Central direct and measured scenarios use 100 assignments. Every displayed  $\pm$  value is an assignment SD, not a standard error or confidence interval. The machine-readable artifacts additionally store numerical Monte Carlo standard error as assignment SD divided by  $\sqrt{n}$ ; it measures precision of the simulated mean only. Survey-design, tariff-response and model-parameter uncertainty are not estimated. Five-assignment mechanism, demographic, poverty-gap, reallocation, duration and sensitivity-grid exercises, and the ten-assignment supply-chain exercise, are exploratory and are excluded from the headline evidence; the take-up grid is scored at the full comparator draw count and reported in the main text.

Reported relative BHC poverty changes hold the poverty line frozen at 60 per cent of the baseline equivalised median; absolute BHC poverty likewise uses 60 per cent of the baseline equivalised median as a fixed line. The cushioning rate is  $1 - \Delta Y / \Delta E$ , where  $\Delta E$  is gross employment-earnings loss and  $\Delta Y$  is household disposable-income loss.

**Thin-support cell.** The OBR three-month duration-equivalent cell is omitted from the main bounds table because its displacement draw rests on 1.2 effective record contributions, with the largest record supplying 91.9 per cent of the loss on average. For completeness:

| Anchor | Months | Margin       | Gross loss<br>(£m) | Exchequer<br>(£m) | Cushioning<br>(%) | Poverty<br>(pp) |
|--------|--------|--------------|--------------------|-------------------|-------------------|-----------------|
| OBR    | 3      | Displacement | $89 \pm 2$         | $35 \pm 8$        | $30.9 \pm 9.0$    | 0.003           |
| OBR    | 3      | Wage cut     | 89                 | 48                | 43.8              | 0.000           |

These rows are sensitivity evidence only, not population estimates.

**Reconciling cushioning and the Exchequer effect.** The Exchequer effect is the change in the aggregate simulated government balance, not the household-side transfer implied by the cushioning identity. The identity implies a household-side offset of  $(1 - \Delta Y/\Delta E) \times \Delta E = \Delta E - \Delta Y$ : for the full-schedule wage cut,  $0.439 \times \pounds 886\text{m} \approx \pounds 389$  million, and for displacement  $0.368 \times \pounds 882\text{m} \approx \pounds 325$  million, against reported Exchequer costs of  $\pounds 484$  million and  $\pounds 387$  million. The wedge is revenue lost outside employee disposable income: employer National Insurance contributions fall with the employee wage bill under every margin, and for displaced workers the zeroing of employee pension contributions and salary-sacrifice arrangements changes taxable pay relative to the gross-earnings accounting in  $\Delta E$ . The government balance also absorbs any other simulated fiscal flows that respond to the shock. Cushion  $\times$  gross therefore understates the Exchequer cost by roughly  $\pounds 87\text{--}95$  million in the central full-schedule scenarios, and the two statistics answer different questions: the cushioning rate measures household income protection per pound of employee earnings lost, while the Exchequer effect measures the total first-round cost to the public finances.

## D Duration scope

The submission design reports three-, six- and twelve-month *duration-equivalent annual stresses*. It scales the probability of a coherent full-year displaced state rather than combining partial-year earnings with full-year unemployment status. These rows preserve expected worker-month loss but are not unemployment-spell simulations. Actual within-year tax, UC, waiting-period and re-employment timing requires a monthly model and is not claimed here.

## E Monthly versus annual UC accounting

The annual model represents an  $m$ -month unemployment spell as a probability  $m/12$  of a coherent full-year displaced state. This appendix bounds the resulting bias with statutory 2026 parameters for a representative single displaced worker aged 25 or over, without children, housing element or capital above the lower limit, at the exposed-sector FRS mean wage of  $\pounds 48,272$ . UC entitlement per worker-month is identical by construction:  $m$  months at the full standard allowance ( $\pounds 424.90$  a month; earnings above the taper-out point in the working months) equals probability  $m/12$  of twelve months. The differences run through income tax and timing. Because the tax schedule is convex, zeroing a full year of earnings forgoes relief at the worker's top marginal slices, whereas a partial-year earner sheds basic- and higher-rate slices: at this wage, the correct partial-year PAYE relief for a three-month spell is  $\pounds 2,414$  against  $\pounds 1,785$  in the annual equivalent, so the annual model *understates* total cushioning for a three- or six-month spell by about 5.2 percentage points (36.5 versus 31.3 per cent) and is exact at twelve months. The five-week UC payment wait shifts timing, not annual entitlement. The primary 12-month contrast in Table 1 is therefore unaffected, and the shorter duration-equivalent rows are conservative for the displacement margin: correcting the annualisation would raise displacement cushioning at those scales and narrow, not widen, the reported gap. Parameters and the full accounting are stored in `results/referee_fixes.json`.